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Monitoring multi-country macroeconomic risk: A quantile factor-augmented vector autoregressive (QFAVAR) approach ☆

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ABSTRACT

A multi-country quantile factor-augmented vector autoregression is proposed to model heterogeneities both across countries and across characteristics of the distributions of macroeconomic time series. The presence of quantile factors enables a parsimonious summary of these two heterogeneities by accounting for dependencies in the cross-sectional dimension as well as across different quantiles of macroeconomic data. Using monthly euro area data, the strong empirical performance of the new model in gauging the impact of global shocks on country-level macroeconomic risks is demonstrated. The short-term tail forecasts of QFAVAR outperform those of FAVARs with symmetric Gaussian errors as well as univariate and multivariate specifications featuring stochastic volatility. Modeling individual quantiles enables scenario analysis of macroeconomic risks, a unique feature absent in FAVARs with stochastic volatility or flexible error distributions.

1. Introduction

The so-called Great Recession of 2008–2009 marked the beginning of an era in which global shocks are more pervasive, are able to generate domino effects to the financial system, and can increase macroeconomic risks to unprecedented levels. Recent evidence about such macroeconomic risks is provided by Adrian et al. (2019) who find that the expected distribution of GDP growth skews left during recessions. This skewness can change over time (Jensen et al., 2020) and is positively related to macroeconomic volatility (Bekaert and Popov, 2019). Similar distributional asymmetries occur for inflation (Korobilis, 2017; López-Salido and Loria, 2019), as for numerous other economic and financial time series. Although some global risks – such as high inflation and oil crises – are eerily familiar to economists from the 1970s, the current unsustainable levels of debt, low growth, and the climate emergency create new challenges for macroeconomic policymakers. In the euro area, which shares common monetary, regulatory, and other policies, measuring and monitoring the heterogeneous performance of countries facing modern global risks is a particularly

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challenging quantitative exercise. As an example, following the first response to the coronavirus (Covid-19) health crisis, euro area GDP was 4.9% below its prepandemic level in the first quarter of 2021. Nevertheless, country-level performance was remarkably heterogeneous, with Ireland reporting growth of 13.2% and Spain experiencing a contraction of −9.3% (Muggenthaler et al., 2021).

In this paper we develop a novel methodology to capture, in a parsimonious and interpretable way, the heterogeneous responses of euro area and country-specific macroeconomic risks to global demand, supply, financial and uncertainty shocks. Our approach builds on combining the strengths of vector autoregressions (VARs) for multivariate structural inference, with the flexibility of modeling individual percentiles of the data distribution of macro data using univariate quantile regression methods. The main challenge with modeling multivariate quantiles is that different percentiles of different variables might be correlated, which is not typically an issue in univariate quantile models.[1] When modeling r quantile levels of n macroeconomic variables a comprehensive approach would require to specify an $(n \times r)$-dimensional VAR that captures all cross-quantile dynamic and static correlations. However, even for moderate values of r and n, such a comprehensive approach to quantile VAR modeling becomes high-dimensional, over-parameterized, and possibly computationally cumbersome. In practice, existing econometric methods oversimplify by specifying n-dimensional VARs that are independently estimated for each of the r quantiles; see for example, recent papers such as Forni et al. (2023), Iacopini et al. (2022), Chavleishvili and Manganelli (2024) and Schüler (2020).

We address this modeling challenge by introducing a quantile factor-augmented vector autoregression (QFAVAR) that extends the widely used factor-augmented VAR method proposed in Bernanke et al. (2005) and Stock and Watson (2005) to the context of quantile regression. We extract variable-specific quantile factors for three percentiles of interest (10th, 50th, and 90th percentiles), in a multivariate setting with several macro variables for several countries. As a result, the factors not only capture common dynamics across the data cross-section (a key feature of multi-country factor studies such as Kose et al. 2003), but they are also a parsimonious way of modeling cross-quantile dependence among the original variables. The key advantage lies in the fact that, akin to recent methodologies presented in Chen et al. (2021) and Korobilis and Schröder (2022), the macroeconomic variables load onto quantile factors independently for each quantile level, ensuring computational simplicity and numerical stability. Beyond this advantage, the QFAVAR facilitates dynamic and contemporaneous correlations among all quantile factors across percentile levels through a joint VAR-state evolution.

Methodologically, the innovative QFAVAR contributes distinctive features to interconnected macroeconometric literatures. First, we draw from existing literature on dynamic factor models (DFMs), where Kose et al. (2003) assess international business cycle synchronization, and Ciccarelli and Mojon (2010) and Mumtaz and Surico (2012) analyze global inflation using common factors. Traditional DFM/FAVAR approaches are limited by the normality assumption in their disturbances, thus, neglecting asymmetries in higher moments in the distribution of macroeconomic data. Even in cases where FAVARs have been extended with flexible nonlinearities and asymmetries,[2] the QFAVAR has the benefit that it is the only modeling approach that can target specific quantiles of interest, enabling interpretable policy-related analysis of tail risks. Furthermore, by integrating quantile factors with VAR dynamics, the QFAVAR extends a recent literature limited to static quantile factors, evident in the work by Ando and Bai (2020), Chen et al. (2021), Clark et al. (2024), Korobilis and Schröder (2022), and Ma et al. (2021). Additionally, the QFAVAR contributes to an emerging literature combining quantile regressions and VAR methods to identify asymmetric effects of macroeconomic shocks, as seen in Castelnuovo and Mori (2022), Forni et al. (2021), and Loria et al. (2019). While existing quantile VAR methodologies may be empirically useful, they often rely on context-specific simplifications.[3] In contrast, the proposed QFAVAR specification is versatile enough to empirically address various topics of policy relevance, such as climate risks, macroeconomic uncertainty, or financial shocks.

Our first contribution is to establish the workings of such a novel specification and to illustrate how it allows for parsimonious VAR inference for quantiles without sacrificing flexibility and generality. In order to achieve this task, we derive a reliable likelihood-based estimator for inference. In particular, we adopt Bayesian methods and modern priors from the statistics and machine learning literature that lead to tuning-free posterior penalized estimation in high dimensions. Numerical approximation of the posterior is based on Markov chain Monte Carlo (MCMC) methods, and in particular a simple generalization of the Gibbs sampler algorithm proposed in Bernanke et al. (2005). By exploiting the state–space form of the QFAVAR, quantile factors and other system matrices can be sampled from their conditional posteriors in a straightforward fashion. We also show that in applications where computational performance is important, such as recursive pseudo out-of-sample forecasting exercises, the Gibbs sampler algorithm can be further simplified. In particular, instead of repeatedly sampling a high dimensional vector of quantile factors one can use a plug-in estimator of the factors, such as those provided by Chen et al. (2021) and Korobilis and Schröder (2022) for the static quantile factor model. This approach is similar to two-step estimation methods typically used in traditional DFMs, where principal components are used as an (asymptotically consistent) approximation to the factors implied by the likelihood function of the DFM (Stock and Watson, 2016).

[1] From another perspective, Adrian et al. (2021) argue that during crises the joint distribution of economic and financial conditions becomes multimodal. Such multimodalities support the argument that percentiles of joint distributions are not symmetrically correlated.

[2] Among many others, papers such as Korobilis (2013), Koop and Korobilis (2014) and Mumtaz and Theodoridis (2017) specify flexible FAVARs with time-varying parameters and/or stochastic volatility; Gorodnichenko and Ng (2017) use a factor model that allows them to explicitly estimate mean and volatility factors.

[3] For example, Bayesian approaches like Mumtaz and Surico (2015) and Schüler (2020) could become computationally cumbersome with the dimensions considered here. Contributions by Ando et al. (2022) are restricted by using observed data for covariance matrices, limiting QVAR estimation to univariate quantile regressions. Chavleishvili and Manganelli (2024) presents a bivariate quantile VAR without interaction between different quantiles. The setting in White et al. (2015), while useful for finance value-at-risk, has limitations for structural macro inference.

Our second contribution is empirical, as we apply the new tool to the problem of assessing the effects of global risks to a series of euro area macroeconomic variables. We collect five variables – namely, inflation, industrial production, the long-term interest rate, an index of economic sentiment, and an index of country-level financial stress – for each of nine euro area countries. We extract quantile factors from each macro variable, by aggregating over all nine countries, such that each quantile factor approximates the distribution of the respective euro area aggregate. For example, the quantile factor extracted from the nine country-level industrial production (IP) series proxies the time series of conditional quantiles of aggregate IP for the euro area. From a measurement perspective these quantile factors are superior in capturing cross-sectional and distributional heterogeneities, compared to fitting univariate quantile regressions directly to the euro area aggregates. By definition, the QFAVAR allows us to augment the unobserved quantile factors with observed variables/factors, and we choose to include global measures of output, supply chain pressures, financial conditions, and economic policy uncertainty.

The empirical facts can be summarized as follows. Quantile factor estimates of the 10th, 50th and 90th percentiles are characterized by evident heterogeneity, implying asymmetries in the distribution of unobserved factors. The QFAVAR is significantly better than a FAVAR with symmetric Gaussian errors in forecasting the left and right tails of the distribution of inflation and IP in the selected euro area countries. Additionally, the QFAVAR performs better out-of-sample compared to key univariate and multivariate models featuring stochastic volatility in a majority of cases. Finally, we illustrate, by means of quantile impulse response functions and quantile forecast error variance decompositions that the QFAVAR captures a large variety of heterogeneities across different quantiles of macroeconomic variables of interest. Imposing a minimal set of sign restrictions, we identify four global shocks and assess their transmission to various quantiles of the euro area data distribution. In the vast majority of cases, we find large asymmetries in the response of macroeconomic data's left and right tails compared to the median response.

This paper unfolds as follows: Section 2 outlines the QFAVAR specification and details Bayesian estimation and inference methods. Section 3 provides a description of the euro area and global data, as well as an evaluation of the in-sample fit of the QFAVAR to this dataset. Section 4 presents forecasting exercises validating the QFAVAR's effectiveness in fitting the data out-of-sample. Section 5 illustrates the effectiveness of the QFAVAR as a structural econometric method for measuring responses of tail risks to key global shocks. Section 6 concludes the paper.

2. Econometric methodology

Our starting point is the factor-augmented vector autoregressive model from Bernanke et al. (2005) and Stock and Watson (2005), adopted for a panel of several macroeconomic time series for several countries. This modeling approach involves extracting a smaller vector of latent factors from the large panel of macroeconomic data. The latent factors evolve jointly with observed variables as a lower-dimensional vector autoregression (VAR). This setting is established in macroeconomics, and the reader should consult (Stock and Watson, 2016) for a thorough review. Here we explain a conceptually straightforward extension of the FAVAR to the quantile setting, show how this extension results in an inherently high-dimensional model, outline how Bayesian inference can help tackle estimation challenges, and, finally, we show how the proposed quantile specification can be deployed for traditional structural VAR analysis.

2.1. A multi-country quantile FAVAR (QFAVAR) specification

Let y_{ijt} denote macroeconomic/financial variable i for country j observed at time t, for $i = 1, \dots, m$, $j = 1, \dots, n$ and $t = 1, \dots, T$. We characterize the distribution of the $mn \times 1$ vector $\boldsymbol{y}_t = \left[y_{11t}, \dots, y_{1nt}, \dots, y_{m1t}, \dots, y_{mnt}\right]'$ by grouping its elements into unobserved, variable-specific factors $f^i_{t,(q)}$ for each quantile level $q = q_1, \dots, q_r$, where $q_l \in (0, 1)$ and $q_{l-1} < q_l$ for $l = 1, \dots, r$. We also assume global-level factors summarized in the $k \times 1$ vector of observed variables $\boldsymbol{g}_t$. The quantile factor model strategy begins by specifying the qth conditional quantile of y_{ijt} as a linear function of the global variables and the variable-specific factor, which is of the form

$$Q_q\left(y_{ijt}|\boldsymbol{g}_t\right) = c_{ij(q)} + \boldsymbol{\gamma}_{ij(q)}\boldsymbol{g}_t + \lambda_{ij(q)} f^i_{t,(q)}, \tag{1}$$

where $c_{ij(q)}$ is a scalar intercept, $\boldsymbol{\gamma}_{ij(q)}$ is a $1 \times k$ vector of loadings on the observed global factors $\boldsymbol{g}_t$, and $\lambda_{ij(q)}$ is the scalar loading (weight) of the scalar, unobserved, variable-specific quantile factor $f^i_{t,(q)}$. The reason for specifying more restrictive, variable-specific factors is examined in detail in Section 2.2.

Following the probabilistic approach in Korobilis and Schröder (2022) this quantile factor model can be represented as a parametric regression of the form

$$y_{ijt} = c_{ij(q)} + \gamma_{ij(q)}\boldsymbol{g}_t + \lambda_{ij(q)} f^i_{t,(q)} + u_{ijt(q)}, \tag{2}$$

where $u_{ijt(q)} \sim AL\left(0, \sigma^2_{ij(q)}, q\right)$ is an asymmetric Laplace disturbance term; that is, it has the functional form

$$f\left(u_{ijt(q)}\right) = \frac{q(1-q)}{\sigma^2_{ij(q)}} \left\{ e^{\left[(1-q)\frac{u_{ijt(q)}}{\sigma^2_{ij(q)}}\right]} \mathbb{I}(u_{ijt(q)} \leq 0) + e^{\left[(-q)\frac{u_{ijt(q)}}{\sigma^2_{ij(q)}}\right]} \mathbb{I}(u_{ijt(q)} > 0) \right\}, \tag{3}$$

with $\mathbb{I}$ denoting the indicator function. Similar to a Bayesian linear regression, where the Gaussian residual is centered around zero, the asymmetric Laplace residual has the qth quantile equal to zero.

An important modeling feature of our approach is that the variable-specific quantile factors and the global factors are contemporaneously and dynamically correlated with each other via a vector autoregressive (VAR) model with p lags. We define the $mr \times 1$ vector $\boldsymbol{F}_t = \left[f^1_{t(q_1)}, \ldots, f^m_{t(q_1)}, f^1_{t(q_2)}, \ldots, f^m_{t(q_2)}, \ldots, f^1_{t(q_r)}, \ldots, f^m_{t(q_r)}\right]'$ that summarizes all unobserved factors at all quantile levels. The VAR(p) that characterizes the joint dynamics of the quantile factors and the global factors is of the form

$$\begin{bmatrix} \boldsymbol{F}_t \\ \boldsymbol{g}_t \end{bmatrix} = \boldsymbol{v} + \boldsymbol{\Phi}_1 \begin{bmatrix} \boldsymbol{F}_{t-1} \\ \boldsymbol{g}_{t-1} \end{bmatrix} + \cdots + \boldsymbol{\Phi}_p \begin{bmatrix} \boldsymbol{F}_{t-p} \\ \boldsymbol{g}_{t-p} \end{bmatrix} + \boldsymbol{\varepsilon}_t, \tag{4}$$

where $\boldsymbol{v}$ is an $l \times 1$ vector of intercept terms, $\boldsymbol{\Phi}_c$ are $l \times l$ matrices of autoregressive coefficients for lagged term $c = 1, \ldots, p$, and $\boldsymbol{\varepsilon}_t \sim N(\boldsymbol{0}, \boldsymbol{\Omega})$ is the disturbance term with $\boldsymbol{\Omega}$ an $l \times l$ full, symmetric and positive definite covariance matrix. Here, $l = mr + k$ is the joint dimension of the vectors $\boldsymbol{F}_t$ and $\boldsymbol{g}_t$. As a result, in contrast to Chen et al. (2021) and Korobilis and Schröder (2022) who estimate factors independently for each quantile, Eq. (4) allows for complex patterns of dynamic correlations among the quantiles to affect the estimation outcomes of the quantile factors. At the same time, as we show in Section 2.4, this latter equation is important in order to perform structural VAR inference using the QFAVAR. Moreover, the model maintains its computational simplicity, as the disturbances $u_{ijt(q)}$ are independent from each other for all i, j, q. Consequently, Eq. (2) represents a series of independent univariate quantile regressions. In contrast, the VAR of the state Eq. (4) may grow substantially, especially when incorporating numerous variable-specific or global factors or considering multiple quantile levels. Nonetheless, addressing high-dimensionality concerns is straightforward by drawing upon a well-established literature on large Bayesian VARs.

2.2. Likelihood and identification

State–space formulation

The QFAVAR consists of seemingly disjointed Eqs. (2) and (4). Without loss of generality, we simplify our notation by dropping intercepts and assuming one lag in the VAR part of the model. Under these simplifications – and as shown in detail in Appendix A – we can combine the two equations of the model into the following linear state–space system

$$\begin{bmatrix} \boldsymbol{Y}_t \\ \boldsymbol{g}_t \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Lambda} & \boldsymbol{\Gamma} \\ \boldsymbol{0} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{F}_t \\ \boldsymbol{g}_t \end{bmatrix} + \begin{bmatrix} \boldsymbol{u}_t \\ \boldsymbol{0} \end{bmatrix}, \tag{5}$$

$$\begin{bmatrix} \boldsymbol{F}_t \\ \boldsymbol{g}_t \end{bmatrix} = \boldsymbol{\Phi} \begin{bmatrix} \boldsymbol{F}_{t-1} \\ \boldsymbol{g}_{t-1} \end{bmatrix} + \boldsymbol{\varepsilon}_t. \tag{6}$$

In this matrix notation, which is derived in detail in Appendix A, $\boldsymbol{Y}_t$ is an $nmr \times 1$ vector with the vector $\boldsymbol{y}_t$ repeated r times; $\boldsymbol{\Lambda}$ is an $nmr \times mr$ block diagonal matrix with the quantile-specific loadings $\lambda_{(q)}$ on its diagonal; $\lambda_{(q)}$ in turn is an nm$\times m$ block-diagonal matrix with the factor-specific loading $\lambda_{i(q)} = \left[\lambda_{i1(q)}, \ldots, \lambda_{in(q)}\right]'$ on its diagonal; and $\boldsymbol{u}_t$ is an $nmr \times 1$ vector of disturbances with qth element $\boldsymbol{u}_{t(q)} = \left[u_{11t(q)}, \ldots, u_{mnt(q)}\right]'$. The above equations define a state–space model that characterizes the joint likelihood of the unobserved state variable $\left[\boldsymbol{F}'_t, \boldsymbol{g}'_t\right]'$ and other latent parameters. Notice that the formulation above stacks all parameters over the cross-sectional dimensions i, j as well as over the quantile levels q_r. Therefore, even for moderate values of m, n, r, the QFAVAR will always be a high-dimensional system with many parameters. For that reason, we partly follow Bernanke et al. (2005) and adopt Bayesian inference as our preferred likelihood-based approach as this will result in regularized posterior estimates and computationally convenient inference.

Factor identification

The interpretation of quantile factors aligns with that of traditional factors. When considering a specific quantile level, which represents a designated portion of the data distribution, a quantile factor is essentially a weighted average of all macroeconomic series measured at that particular quantile level. Consequently, as with any other factor model, the QFAVAR suffers from a lack of identification due to the presence of the product of two latent parameters in Eq. (5), namely the factors $\boldsymbol{F}_t$ and the loadings $\boldsymbol{\Lambda}$. While the common component $\boldsymbol{\Lambda}\boldsymbol{F}_t$ is identified, separating $\boldsymbol{\Lambda}$ from $\boldsymbol{F}_t$ involves additional restrictions. First, note that due to the quantile structure of the measurement equation of the QFAVAR, the matrix $\boldsymbol{\Lambda}$ is block-diagonal with element $\boldsymbol{\Lambda}_q$, as shown in detail in Appendix A. Also, due to the fact that we extract variable-specific factors, we only allow the same macroeconomic variable i for all countries to load on a factor $f^i_{t,(q)}$, and not into other factors. That is, loading $\lambda_{ij(q)} = 0$ for factor $f^k_{t,(q)}$ whenever $i \neq k$, meaning that variable y_{ijt} does not load on the kth variable quantile factor. This structure in the way factors are extracted implies that each submatrix $\boldsymbol{\Lambda}_q$ has a block diagonal structure with several zero elements naturally occurring. Consequently, as is the case with many multi-country factor models estimated in the literature, the panel structure in the data provides a lot of zero restrictions that help identify the factors more easily. Following this existing multi-country literature, we simply normalize the qth loading of each quantile factor to be one so that we can identify the sign and scale of each factor; see the discussion in Kose et al. (2003, Section I).

Number of factors

The number of factors in our empirical application is fixed, as the restriction to extract variable-specific factors means that we have as many quantile factors as groups of macroeconomic variables. Is such a panel restriction justified? An alternative approach would be to allow all macroeconomic variables to load on a large number of factors without any restrictions, and then use an information criterion (Bai and Ng, 2002) to determine the optimal number of factors. Unfortunately, for several reasons, the traditional approach to treating factors is not recommended for the multi-country quantile framework we are interested in. These reasons are the following:

1. **Statistical identification of factors:** In a likelihood-based setting, if zero restrictions are not provided by the panel structure of the dataset, they have to be imposed in some other, arbitrary way. For example, Bernanke et al. (2005) impose the upper block of the loadings matrix to be the identity matrix, implying that the ordering of variables in the dataset affects factor estimation. Such ordering restrictions can be detrimental for factor estimation and they are not welcome.
2. **Economic identification of factors:** Since factors are latent variables, estimating them without any constraints results in a loss of economic interpretability within the model. Variable-specific factors have a clear interpretation as they simply approximate the aggregate quantities for the whole of the euro area. For instance, the industrial production quantile factor is a linear combination of all individual country industrial production series; hence, it can be considered a proxy for the total industrial production of the euro area. This identification is economically important, as the quantile factors enter the VAR specification, and that way structural shocks to the quantile factors can be interpreted as structural shocks to quantiles of euro area aggregates. This issue is underscored and explored in detail in Figure S3 of the Online Supplement.
3. **Interpretability of quantile estimates:** The interpretation of a 10th percentile factor from all macro variables of all countries in our dataset is far from intuitive, as this would represent a weighted average of the 10th percentiles of variables with diverse distributional characteristics. As demonstrated by Chen et al. (2021), it is numerically feasible to estimate such quantile factors from all available data, and these factors also provide a good fit. This is because these authors, in their simulation setup, assume that factors are normally distributed, and only idiosyncratic errors have flexible distributions. If this is indeed the true data data-generating process (DGP), then extracting unrestricted quantile factors from all variables in the dataset will indeed work well.
However, what if we suspect that the underlying factor structure in the data is not normally and symmetrically distributed? In this case, we have a factor $f_t \sim \mathscr{P}$ with $\mathscr{P}$ a flexible, asymmetric distribution. An estimate of the qth quantile factor, $\widehat{f}_{t,(q)}$, will not necessarily be equivalent to the true qth quantile of the distribution $\mathscr{P}$. To see this, consider a single true factor $\boldsymbol{f}$ extracted from two positively correlated variables $\boldsymbol{y}$. If the loading in one variable is positive, and the loading on the other variable is negative, then the left tail (e.g. 10th percentile) of the true factor is a combination of information in the left tail of the first variable and the right tail of the second variable (due to the negative loading). In this case, the quantile factor estimate $\widehat{f}_{t,(10\%)}$ will wrongly utilize information in the 10th percentiles of both variables. If the true factor is not normal (hence, there is no symmetry between the 10th and 90th percentiles), quantile factor estimation will collapse. We illustrate this point more clearly in Section S5 of the Online Supplement by means of a numerical example where we simulate a factor from an asymmetric distribution.
By extracting factors in blocks of the same variable for different countries, we ensure that the quantile factors do not suffer from the numerical issue described above. For example, the 10th quantile inflation factor is simply a linear combination of the 10th quantiles of inflation series for each country, and these country-level inflation series are always positively correlated with each other. One might argue that our model is a restricted one and might not be well-supported empirically, as it imposes lots of zero restrictions in the loadings matrix compared to extracting factors from all variables without imposing restrictions. We indeed show, also in Section S5 of the Online Supplement, that our variable-specific factors provide far better out-of-sample fit compared to QFAVARs estimated with three, five or seven quantile factors extracted unrestrictedly from all macro data $\boldsymbol{y}_t$.

To shed further light onto the quality of the factor estimates in this block diagonal setting, Online Supplement Section S3.2 presents a thorough simulation study in which we compare the quality of the factor estimates from the QFAVAR with the estimates obtained with the static quantile factor model in Chen et al. (2021) for different DGPs. In all cases, the QFAVAR yields more accurate factor estimates especially for the tail factors. This suggests that accounting for the joint dynamics of the quantile factors in a VAR yields empirically relevant performance gains also in this seemingly restrictive setting.

In other applications, it might be useful to estimate the number of factors using information criteria. Therefore, Online Supplement Section S3.1 provides extensive simulations that study the accuracy of the information criteria in Bai and Ng (2002) in the context of the QFAVAR. We find that especially the information criteria IC_1, IC_2, PC_1, and PC_2 yield accurate estimates of the correct number of factors already for moderate sample sizes.

2.3. Prior distributions and posterior inference

Shrinkage priors for flexible inference on loadings and lag length

The first reason for addressing estimation using Bayesian inference is the vast availability of suitable prior distributions that provide automatic regularization of the joint likelihood. This is particularly advantageous in cases where only a limited number of observations is accessible, such as the estimation of extreme quantiles. Following Feldkircher et al. (2022), Korobilis (2022) and others, we specify the horseshoe prior for sparse signals of Carvalho et al. (2010). This prior is specified for the elements of the matrices $\boldsymbol{\Lambda}$ and $\boldsymbol{\Gamma}$, as well as for the elements of the VAR coefficients $\boldsymbol{\Phi}$. For a generic b-dimensional vector of parameters $\boldsymbol{\theta}$ (where $\boldsymbol{\theta}$ represents column vectors obtained from vectorizing the parameter matrices $\boldsymbol{\Lambda}, \boldsymbol{\Gamma}, \boldsymbol{\Phi}$, respectively) the Horseshoe prior takes the form

$$\boldsymbol{\theta}|\xi, \boldsymbol{\eta} \sim \prod_{i=1}^{b} N\left(0, \xi\eta_i\right), \tag{7}$$

$$\xi \sim C^{+}(0,1), \tag{8}$$

$$\eta_i \sim C^{+}(0,1). \tag{9}$$

Shrinkage estimators introduce regularization to an equivalent unrestricted estimator through a scalar factor κ, determining the extent to which the unrestricted estimator is pulled toward zero.[4] In the case of the prior above, the implied factor κ is $Beta(0.5, 0.5)$ distributed, that is, it has the shape of a horseshoe. In turn, this means that with smaller parameter spaces the posterior under a horseshoe prior will tend to be unrestricted, but as the parameter space increases relative to the number of observations, an increasingly larger amount of shrinkage towards zero is favored. These properties make the horseshoe prior ideal for our large dimensional parameter space. Both theoretically and empirically the horseshoe performs well and it is a default automatic choice for many researchers; see the detailed review of this literature in Korobilis and Shimizu (2022). Conjugate inverse gamma priors are specified for the scalar asymmetric Laplace scale coefficients, denoted as $\sigma_{ij(q)}$. Finally, the VAR covariance matrix $\boldsymbol{\Omega}$ is assigned an inverse Wishart prior.

Conditional posteriors and Markov-chain Monte Carlo approximation

Other than regularized estimation via prior distributions, the second reason for choosing a Bayesian approach to inference, is computational convenience. The state–space model of Eqs. (5)–(6) is linear but non-Gaussian because $\boldsymbol{u}_t \sim \prod_{ijq} AL(0, \sigma_{ij(q)}, q)$. Additionally, the presence of the quantile common component $\boldsymbol{\Lambda F}_t$, which is a product of two high-dimensional unobserved quantities, complicates state–space estimation further. However, estimation via the Gibbs sampler simplifies inference, because conditional distributions in the QFAVAR have a simple form. The asymmetric Laplace distribution can be written as a mixture of Gaussian distributions, converting the linear state–space model into conditionally normal form. The parameters $\boldsymbol{\Lambda}$ and $\boldsymbol{F}_t$ can be sampled one at a time, conditional on one other.[5] In practice, our estimation strategy combines established ergodic samplers, and can be outlined in the following steps:

1. Sample $\left[\boldsymbol{F}_t', \boldsymbol{g}_t'\right]'$ conditional on values of all other system matrices from the state–space model of Eqs. (5)–(6). This step can be implemented using numerous approaches proposed in the literature, most notably the simulation smoother of Carter and Kohn (1994).
2. Sample $\lambda_{ij(q)}, \gamma_{ij(q)}, \sigma_{ij(q)}$ (and c_{ij}, if an intercept is present) conditional on $f^{i}_{t(q)}$, for each i, j, q, using Eq. (2). This is simply a univariate quantile regression, and simple conditional posteriors are provided in Khare and Hobert (2012).
3. Sample $\boldsymbol{\Phi}_1, \ldots, \boldsymbol{\Phi}_p, \boldsymbol{\Omega}$ (and $\boldsymbol{v}$, if an intercept is present), conditional on all quantile factors $\boldsymbol{F}_t$ using Eq. (4). For numerical stability of forecasts and other projections, we only accept posterior samples of $\boldsymbol{\Phi}_1, \ldots, \boldsymbol{\Phi}_p$ that imply a stationary VAR process, while rejecting explosive samples. This is a simple Bayesian VAR, and posterior conditionals are also quite standard, see Koop and Korobilis (2010).

Of course, in steps 2 and 3 one needs to account for the use of the Horseshoe hierarchical prior, but this can be trivially incorporated using the hierarchical formulation of this prior proposed in Makalic and Schmidt (2016). The outcome is a Gibbs sampler that closely resembles the sampler in Bernanke et al. (2005) for the FAVAR model, making it straightforward and user-friendly in structure.

2.4. VAR inference in the QFAVAR

The QFAVAR implies a joint VAR for the r quantiles of country-level macroeconomic variables $\boldsymbol{y}_t$ and the mean (expectation) of the global variables $\boldsymbol{g}_t$. Here we follow Bernanke et al. (2005), and Stock and Watson (2005) and we assume that the idiosyncratic disturbances $\boldsymbol{u}_t$ in Eq. (5) are not relevant for structural inference. This assumption allows to implement VAR inference in two steps: first, implement all regular exercises of interest (forecasting, impulse response analysis, historical decompositions) using the mean VAR in Eq. (6), and then project these quantities into the quantiles of the original variables $\boldsymbol{y}_t$ using the projection matrices $\boldsymbol{\Lambda}$ and $\boldsymbol{\Gamma}$ in Eq. (5). We explain here how this is done for forecasting and for structural inference.

Prediction

Using standard formulas (Kilian and Lütkepohl, 2017) h-step-ahead forecasts of $\left[\boldsymbol{F}_{t+h|t}', \boldsymbol{g}_{t+h|t}'\right]'$, for some integer $h > 0$, can be obtained from the VAR in Eq. (6) as follows

$$\begin{bmatrix} \boldsymbol{F}_{t+h|t} \\ \boldsymbol{g}_{t+h|t} \end{bmatrix} = \boldsymbol{\Phi}^{h} \begin{bmatrix} \boldsymbol{F}_{t} \\ \boldsymbol{g}_{t} \end{bmatrix}. \tag{10}$$

Even if the VAR state equation has more than one lag, it is always feasible to write it in appropriate companion form so that Eq. (10) is applicable. If the VAR has an intercept, then the formula needs to be adapted accordingly. All such extensions are standard and are provided in Kilian and Lütkepohl (2017), so we do not provide them here for the sake of simplicity in notation. The VAR provides forecasts of the vector of quantile factors $\boldsymbol{F}_t$ and the global factors $\boldsymbol{g}_t$. In turn, these forecasts are projected to the quantiles of the original variables using the formula

$$\boldsymbol{Y}_{t+h|t} = [\boldsymbol{\Lambda}, \boldsymbol{\Gamma}] \left[\boldsymbol{F}_{t+h|t}', \boldsymbol{g}_{t+h|t}'\right]'. \tag{11}$$

[4] In this scenario, the unrestricted estimator arises from imposing a normal prior on θ with infinite variance.

[5] While this approach induces high correlation of consecutive Monte Carlo samples of these parameters, this issue can be easily alleviated by storing only every nth posterior sample, for an appropriate choice of n.

The $nmr \times 1$ vector $\boldsymbol{Y}_{t+h|t}$ stacks the forecasts of all quantile levels of euro area macro series $\boldsymbol{y}_t$. That is, the first nm elements of $\boldsymbol{Y}_{t+h|t}$ correspond to forecasts $\boldsymbol{y}_{t+h|t,(q_1)}$, while the last nm elements of the same vector correspond to forecasts of $\boldsymbol{y}_{t+h|t,(q_r)}$. Finally, notice that in a Bayesian setting the formulas above are computed for each sample of all parameters $\boldsymbol{\Phi}, \boldsymbol{\Sigma}, \boldsymbol{\Lambda}, \boldsymbol{\Gamma}$, such that the full posterior predictive density is obtained. We then obtain the posterior mean of the posterior predictive density which is the Monte Carlo mean of all forecasts for quantile level q, that is

$$\mathbb{E}\left(\boldsymbol{y}_{t+h|t,(q)}\right) = \frac{1}{S}\sum_{s=1}^{S}\left[\boldsymbol{\Lambda}^{(s)}, \boldsymbol{\Gamma}^{(s)}\right]\left(\boldsymbol{\Phi}^{h}\right)^{(s)}\begin{bmatrix} \boldsymbol{F}_t^{(s)} \\ \boldsymbol{g}_t \end{bmatrix}. \tag{12}$$

S, for $s = 1, \ldots, S$, denotes the total number of samples obtained from the parameter posterior distributions, and superscript (s) denotes the individual conditional posterior sample for each unknown parameter. Therefore, the Monte Carlo posterior mean $\mathbb{E}\left(\boldsymbol{y}_{t+h|t,(q)}\right)$ is our final interpretation of the qth quantile forecast of the euro area macroeconomic variables.

There is an important practical issue worth mentioning at this point. QFAVAR quantile forecasts might cross, meaning that forecasted conditional quantile values might not necessarily be ordered as we move from one quantile to the next. This is a known issue with many existing quantile regression methods. However, there are at least three options for ensuring monotonicity of quantile estimates. The first solution is to reject posterior samples that do not comply with monotonic solutions. The second is to use a posterior smoothing procedure that ensures monotonicity of posterior mean estimates of quantiles (see Rodrigues and Fan, 2017). The last option is to fit the conditional quantile estimates on a flexible distribution (either parametric or nonparametric), and then to infer the implied forecasted quantiles based on the estimate of this distribution. None of these three solutions/procedures comes without a drawback. The first solution to the quantile crossing problem can become computationally cumbersome when many samples are rejected as inappropriate. The second solution introduces some bias in the final conditional quantile estimates. Finally, the third solution makes out-of-sample predictions dependent on the choice of the distribution and its hyperparameters (or degree of smoothing, in the nonparametric case). In our empirical application we focus on forecasts of three distinct quantiles, namely 10th, 50th and 90th percentiles so crossing is never an issue. In this case, we always quote raw quantile forecasts given by Eq. (12), which also helps replicability of our results compared to post-processed quantile forecasts.

Structural inference

Structural inference in the QFAVAR closely mirrors forecasting procedures. Initially, the task involves mapping the reduced-form VAR disturbances ϵ_t to the structural shocks implied by a structural VAR form. In theory, any structural identification scheme available in the literature, ranging from short and long restrictions to sign restrictions and identification using proxies (see Kilian and Lütkepohl, 2017, for a comprehensive review), can be applied to the QFAVAR.

However, the specific structure of the QFAVAR, where different quantiles of the same factors/variables manifest as endogenous variables, poses challenges. This structure implies that, for instance, a naive recursive identification scheme does not make sense as it imposes a causal ordering on different quantiles of each variable. At the same time, the presence of quantile factors in the QFAVAR allows for the exploration of potential asymmetries in the transmission of global shocks. Economic theory implicitly posits that the effects of a truly structural shock should be identical throughout the distribution of possible outcomes of a macroeconomic variable. In practice, identifying truly structural shocks at the global dimension is challenging, making it empirically relevant to consider possible asymmetries and sign reversals in the transmission of global shocks to different quantiles of macroeconomic variables.

To unveil these intriguing patterns, we adopt a strategy where we initially derive generalized impulse response functions (GIRFs). While GIRFs lack structural interpretation, they impose a minimal set of restrictions compared to, for example, recursive identification, and they "let the data speak". In a subsequent step, we leverage the information in the GIRFs to motivate a set of sign, zero, and magnitude restrictions applied solely to the median factor. These restrictions on the median are adequate for identifying global shocks, while simultaneously leaving the response of the 10th and 90th percentile factors unrestricted. This approach enables the QFAVAR to uncover notable asymmetries across the distribution of macroeconomic data and their cross-section. Further details on our approach are discussed in Section 5.

3. Data and in-sample model fit

3.1. Euro area macroeconomic variables and global data

We use five macroeconomic variables from nine euro area countries observed over the sample 1996M1–2022M12. The countries, series, their acronyms, and their sources, are shown in the top panel of Table 1. Three series come from the Statistical Data Warehouse (SDW) of the European Central Bank. Industrial production is from the OECD data website, and the Economic Sentiment Index (ESI) is a composite index maintained by the DG ECFIN (and accessible from the website of Eurostat, the main statistical agency in Europe). All series were accessed in March 2023. Consumer prices are not seasonally adjusted, so we convert these to year-on-year growth rates using the transformation $100(\log P_{t+12} - \log P_t)$; we use the same annual growth transformation for industrial production and the ESI in order to create smoother series, despite the fact that these series are provided seasonally adjusted. Finally, LTIR and CLIFS are left observed in levels.

The global series include Global Economic Conditions (GECON), the Global Supply Chain Pressure Index (GSCPI), the Financial Conditions Index (FCI), and Global Economic Policy Uncertainty (GEPU). The global economic conditions (GECON) is constructed in Baumeister et al. (2022) as a factor from 16 key global series, and it is a proxy for global output/production. The supply chain pressure index is constructed by the Federal Reserve Bank of New York, and their "[...] research indicates that changes in the

Table 1
Euro area and global variables.

Countries	Macroeconomic variables	Acronym	Source
Austria (AT)	Harmonized Index of Consumer Prices – Overall index	HICP	SDW[a]
Belgium (BE)	Industrial Production Index – Total index	IP	OECD[b]
Germany (DE)	Long Term Interest Rate	LTIR	SDW[a]
Spain (ES)	Economic Sentiment Index	ESI	Eurostat[c]
Finland (FI)	Country-Level Index of Financial Stress	CLIFS	SDW[a]
France (FR)			
Italy (IT)			
Netherlands (NL)			
Portugal (PT)			
	Global variables	**Acronym**	**Source**
	Global Economic Conditions Index	GECON	Baumeister et al. (2022)[d]
	Global Supply Chain Pressure Index	GSCPI	NY Fed[e]
	Chicago Fed National Financial Conditions Index	FCI	Chicago Fed[f]
	Global Economic Policy Uncertainty	GEPU	EPU webpage[g]

[a] https://sdw.ecb.europa.eu/
[b] https://data.oecd.org/industry/industrial-production.htm
[c] https://ec.europa.eu/eurostat/databrowser/view/teibs010/default/table?lang=en
[d] https://sites.google.com/site/cjsbaumeister/research
[e] https://www.newyorkfed.org/research/policy/gscpi#/overview
[f] https://www.chicagofed.org/research/data/nfci/current-data
[g] https://www.policyuncertainty.com/

GSCPI are associated with goods and producer price inflation in the United States and the euro area".[6] The FCI is the national index for the United States, produced by the Chicago Federal Reserve Bank. It is not, strictly speaking, a global average of multiple countries, as are the other three global factors. Nevertheless, this FCI is a weighted average of over 100 financial time series covering developments in stock, foreign exchange, bond, and other key US markets at the forefront of global financial activity. Therefore, for euro area countries in particular, the U.S. FCI can be an excellent proxy of the global financial environment. Finally, the GEPU index has been a very popular series and does exactly what its name suggests, that is, proxies global economic policy uncertainty. All variables are used in their original index form. Plots of the global series are provided in the Online Supplement, while sources and detailed definitions can be found in the bottom panel of Table 1.

3.2. Quantile factor estimates

It should come as no surprise that common drivers characterize a large portion of the euro area's macroeconomic dynamics. Other than the conceptual reasons for extracting variable-specific factors, discussed in the previous Section, Figure S2 of the Online Supplement presents additional evidence that supports extracting factors by variable: euro area data have a block-diagonal correlation structure. This means the same variables for different countries are strongly correlated, while different variables for the same country are at best weakly correlated. Thus, it is not surprising that five variable-specific factors would sufficiently capture most of the variability in the macro data.

Next, it is important to demonstrate that significant dynamics in the quantiles of macroeconomic variables remain hidden when only mean estimates of factors are considered. Furthermore, these dynamics are not only interesting in and of themselves, but they also imply significant heterogeneity and asymmetries in the transmission of economic shocks. Fig. 1 depicts the estimated QFAVAR factors using the MCMC estimator. Quantile factor estimates plotted are defined as posterior means of the 10th, 50th, and 90th percentile factors. For the sake of clarity the mean (FAVAR) factors are not plotted in this figure, as these are fairly indistinguishable from the median (50th percentile) factors. Therefore, median factors are a reference point when comparing those to tail factors. In all five time series there is evident heterogeneity in the dynamics of the tail factors versus the median factor. Certain periods stand out, most notably the large hump in the 90th percentile factor of long-term interest rates around the period of the eurozone debt crisis, that saw sovereign bond spreads for peripheral countries skyrocket.

3.3. Computational and numerical considerations

Finally, we examine how accurate factor estimates from the QFAVAR are. The Online Supplement provides several exercises using simulated data experiments, where it is established that the model recovers simulated quantile factors quite accurately. Next, a question of interest is whether the MCMC algorithm is efficient enough in producing reliable samples from the true parameter posterior. Convergence diagnostics, also provided in the Online Supplement, support the good properties of the Monte Carlo sampling procedure. The QFAVAR is a conditionally Gaussian state–space model and several efficient simulation methods for such models exist and their properties are well understood.

[6] See more details on the NY Fed webpage https://www.newyorkfed.org/research/policy/gscpi#/overview (accessed 23 November 2023).

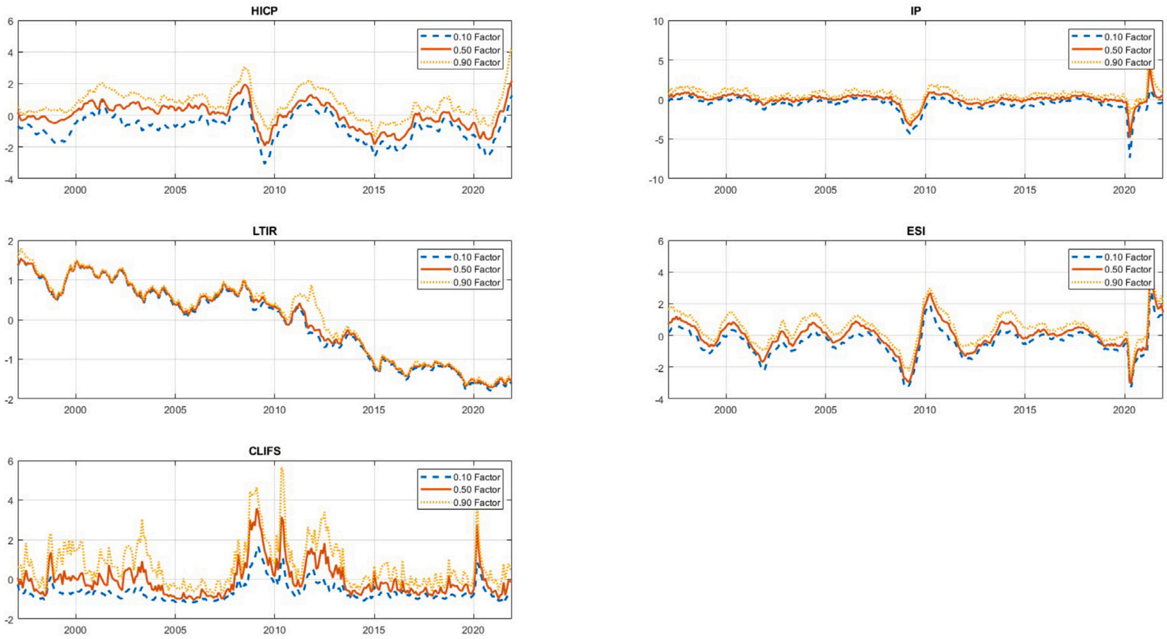

Fig. 1. Markov Chain Monte Carlo estimates (posterior means) of the five euro area quantile factors (10th, 50th, and 90th percentiles).

However, a note of caution should be made about two important aspects of the QFAVAR. First, as with all factor models, sampling from the posterior of the loadings conditional on the factors and vice versa, causes a Markov chain that is quite correlated. In order to reduce this correlation, we only retain one in every 100 samples we draw from these conditional posteriors. Second, the QFAVAR implies a state equation that is a VAR of high dimensions. In our application we deal with a 19-variable VAR, but should we decide to include, for example, four more quantile levels in our analysis, the state equation becomes a 39-dimensional VAR. While methods for sampling from such large Bayesian VARs exist, a challenging task is to ensure that the sampled parameters imply a stable VAR process. Given that this VAR has to be built using different estimates of the quantile factors in each MCMC iteration, sampling stable autoregressive matrices $\boldsymbol{\Phi}$ is not warranted.[7]

For these two reasons, in certain cases it might be better to follow the traditional DFM/FAVAR literature and simplify the one-step estimation procedure into a two-step approximate procedure. Stock and Watson (2005), among numerous others, argue that when the number of macroeconomic series is large, the factors of a DFM can be approximated using principal components. Using the principal components as plug-in estimators of the factors, any remaining parameters can be estimated with OLS. Bernanke et al. (2005) use such a two-step procedure for the FAVAR, alongside their proposed one-step MCMC estimation algorithm. In the case of the QFAVAR, we have similar quantile factor estimates readily provided by static quantile factor models. Chen et al. (2021) develop a simple, iterative, loss-based estimation procedure, and Korobilis and Schröder (2022) develop a fast likelihood-based estimation algorithm based on variational Bayes methods. Fig. 2 shows quantile factor estimates from our dataset using the latter estimation procedure. While numerical estimates are not identical to those presented in Fig. 1, they are quite similar.

The quantile factor estimates using the algorithm in Korobilis and Schröder (2022) only require seconds of computational time. One can use these as plug-in estimates of the quantile factors and then run our MCMC algorithm for the QFAVAR conditional on these. Doing so, means that there is no sampling uncertainty regarding factors; sampling of the loadings becomes very efficient as their conditional posterior does not depend on the factor any more; and sampling a stable VAR also becomes easier, as this VAR is estimated for each MCMC iteration using the same plug-in quantile factor estimates. Given these computational considerations, in the next Section we adopt the two-step MCMC algorithm, and we only use the one-step algorithm for structural VAR inference.

4. Out-of-sample evaluation of tail risks

We perceive the QFAVAR model primarily as a suitable tool for structural econometric analysis due to its capacity to target specific quantiles of interest while maintaining simplicity and interpretability. Nonetheless, the efficacy of an econometric model ultimately hinges on the reliability of its out-of-sample forecasts and projections. This section evaluates the QFAVAR's ability to generate reliable quantile forecasts and compares it to key benchmarks found in the econometric literature. To better understand

[7] In practice, we aim to only retain samples of $\boldsymbol{\Phi}$ that imply a stationary VAR process, however, such accept/reject algorithms are notoriously computationally inefficient in the Bayesian VAR literature, especially in larger dimensions.

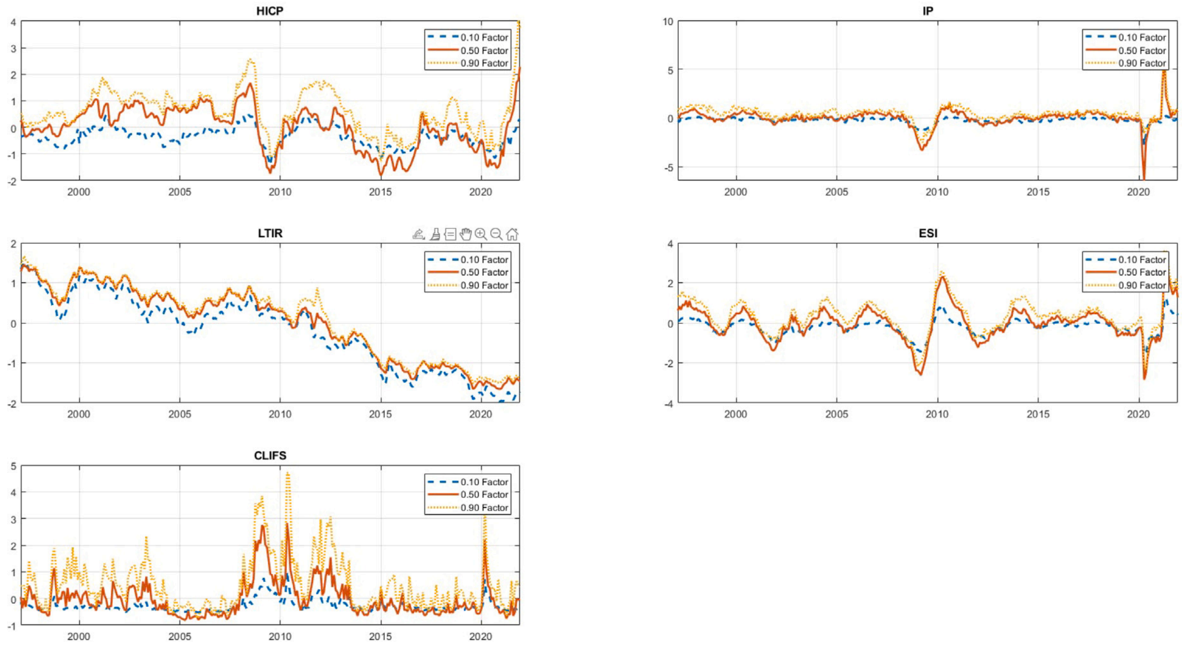

Fig. 2. Variational Bayes estimates (posterior means) of the five euro area quantile factors (10th, 50th, and 90th percentiles).

the origins of the QFAVAR's advantages, we divide this section into three separate forecast comparisons. Doing so, allows to gradually pinpoint the precise contributions of our model to existing stylized facts documented in the macroeconomic forecasting literature. First, we compare the QFAVAR to the FAVAR in order to find out whether our quantile extension is worth pursuing compared to the simpler model of Bernanke et al. (2005). Next, following the debate in Plagborg-Møller et al. (2020) we attempt to find whether global predictors enhance the predictability of extreme quantiles by comparing the QFAVAR with a restricted version of the same model without global predictors. Finally, we contribute to the recent debate in Clark et al. (2024) and others, who argue that models featuring Gaussian errors with stochastic volatility are as good at tail forecasting as (univariate) quantile regressions.

Throughout this section we focus exclusively on forecasting either the left or right tail of inflation and industrial production. Admittedly, evaluating only a certain area (left or right tail) of the distribution of inflation and output is of paramount importance to policy-makers who are interested in assessing worst-case scenarios for these two variables of interest.[8] We set up a straightforward recursive pseudo-out-of-sample (poos) exercise where we begin the estimation with 50% of the total sample, forecast one to 24 months ahead, add one more observation at the end of the sample, estimate the models, and forecast again up to 24 horizons ahead. We follow Manzan (2015) and evaluate the tail forecasting performance using the following quantile score function

$$QS_{t|t-h}(q)^{M} = \left[y_{ijt} - Q_q(y_{ijt})\right]\left[\mathbb{I}\left(y_{ijt} \leq Q_q(y_{ijt})\right) - q\right]. \tag{13}$$

This score function is evaluated for the QFAVAR and any number of competing models, that is $M = \{QFAVAR, competing_1, competing_2, \dots\}$; the two extreme quantile levels $q = 0.1, 0.9$; four forecast horizons $h = 1, 6, 12, 24$; two variables of interest $i = HICP, IP$; and all nine euro area countries j. The QS function is a piecewise linear asymmetric loss function; therefore lower values signify better performance. Unless otherwise stated, the QFAVAR as well as all competing autoregressive models are estimated using $p = 2$ lags. Following the discussion of the previous section, the QFAVAR is estimated using the faster and simpler two-step MCMC estimation algorithm.

4.1. Is the QFAVAR better than the FAVAR at capturing tail risks?

In this subsection we shed more light onto the empirical fit of the QFAVAR, compared to the FAVAR model, which is an established benchmark in the literature. Because we are interested in comparing the performance of only two different classes of models, we again follow Manzan (2015) and calculate the t-statistic testing the equality of the average of the models' QS functions over the whole out-of-sample period. This is defined as the spread between the QS values of the QFAVAR versus the QS values of the FAVAR. Values of this statistic for different variables and forecast horizons are shown in Table 2. The visibly higher proportion

[8] Nevertheless, many times the full forecast distribution of the variables of interest is required, most notably when preparing fan charts similar to the ones maintained by the Bank of England (Britton et al., 1998). In such cases, one has to construct the full forecast distribution by interpolating forecasts of a wide range of quantiles. In Section S6.2 of the Online Supplement we provide comprehensive instructions on employing the QFAVAR for generating density forecasts.

Table 2
T-statistics of equal quantile predictive accuracy.

	$t-stat_{10}$				$t-stat_{90}$			
	$h=1$	$h=6$	$h=12$	$h=24$	$h=1$	$h=6$	$h=12$	$h=24$
HICP.AT	**−6.48**	**−6.82**	**−6.92**	**−4.51**	**−3.53**	**−3.64**	**−4.30**	−1.87
HICP.BE	**−4.83**	**−7.36**	**−4.92**	**−4.04**	**−3.71**	**−3.12**	**−3.16**	−1.42
HICP.DE	**−5.62**	**−6.40**	**−5.34**	**−3.51**	2.37	0.94	−0.83	−1.23
HICP.ES	**−7.70**	**−7.23**	**−4.99**	**−3.63**	**−4.02**	**−2.69**	**−2.39**	−1.86
HICP.FI	**−7.88**	**−7.00**	**−8.40**	**−5.33**	**−5.21**	**−4.58**	**−4.20**	**−2.30**
HICP.FR	**−6.32**	**−7.95**	**−5.88**	**−3.63**	**−4.64**	**−2.79**	**−3.65**	−1.83
HICP.IT	**−9.14**	**−8.16**	**−6.29**	**−4.45**	−1.57	**−2.84**	**−2.59**	−1.05
HICP.NL	**−10.54**	**−10.44**	**−8.82**	**−5.86**	**−3.39**	**−5.99**	**−4.63**	−1.49
HICP.PT	**−6.06**	**−7.27**	**−6.65**	**−8.08**	**−4.01**	**−4.81**	**−4.09**	−1.27
IP.AT	**−2.75**	**−2.34**	**−3.38**	**−4.55**	**−5.30**	**−2.50**	**−2.60**	**−2.17**
IP.BE	**−5.01**	**−4.55**	**−4.16**	**−5.88**	**−4.06**	**−2.95**	**−3.38**	**−2.36**
IP.DE	**−3.39**	−1.96	**−2.41**	**−3.74**	**−5.76**	**−2.76**	**−3.11**	−1.29
IP.ES	**−3.18**	−1.65	**−2.55**	**−3.01**	**−4.77**	**−2.16**	**−2.25**	−1.90
IP.FI	**−5.80**	**−4.93**	**−5.79**	**−6.29**	**−6.09**	**−4.09**	**−5.70**	**−3.92**
IP.FR	−1.95	−0.57	−1.23	**−2.06**	**−5.16**	**−2.26**	**−2.35**	−1.30
IP.IT	−1.56	−0.96	−1.43	**−2.00**	**−3.27**	−1.77	−1.90	−1.06
IP.NL	**−6.20**	**−5.63**	**−6.14**	**−6.91**	**−6.42**	**−4.11**	**−4.89**	**−3.91**
IP.PT	**−4.19**	**−2.12**	**−2.66**	**−3.56**	**−5.60**	**−3.55**	**−3.78**	**−4.04**

Notes. Entries in this table are t-statistic values for the null hypothesis of equal accuracy of the quantile forecasts from the QFAVAR relative to the FAVAR benchmark. Columns 2-5 show the statistics for 10th percentile forecasts over forecast horizons $h = 1, 6, 12, 24$ months, and columns 6-9 show the statistics for the 90th percentile forecasts over the same forecast horizons. Because almost all values of the statistic are negative, this shows that the QFAVAR experiences less forecast loss than the benchmark FAVAR. Values lower than the critical value of −1.96 indicate that QFAVAR tail forecasts are significantly better than the FAVAR forecasts at the 5% level (indicated by bold numbers).

of negative values implies that, overall, the QFAVAR experiences less forecast performance loss in both the left and right tails of the distribution of HICP and IP. For HICP, performance gains are statistically significant at the 5% level for all forecast horizons and countries for the bottom tail of the distribution. For the top tail, the forecast performance gains of the QFAVAR are significant for $h = 1, 6, 12$ and all countries except Germany. For $h = 24$ forecast performance gains are only significant for Finland. For industrial production the picture is generally comparable. For the bottom tail of the distribution, the performance gains are significant at all forecast horizons and for most countries. Notable exceptions are France and Italy for which the forecast performance gains are only significant for $h = 24$. For the top tail, the performance gains are significant for $h = 1, 6, 12$ and all countries except Italy, where gains are only significant at $h = 1$. In addition, significant gains emerge for $h = 24$ for Austria, Belgium, Finland, the Netherlands, and Portugal. Generally, the QFAVAR provides clear performance gains over the symmetric, Gaussian disturbances of the FAVAR, highlighting the importance of flexible asymmetric modeling of conditional quantiles within a multivariate setting.

4.2. Are global variables relevant for forecasting tail risk?

With multiple global shocks affecting the euro area, it is unclear how these transmit to country-level risks of inflation and industrial production (Panetta, 2023). Understanding the underlying transmission channels of such global predictors is an important issue for policy-makers. A good starting point towards this direction is the related debate on whether financial predictors affect the left quantile of GDP. Despite the fact that Adrian et al. (2019) find that financial conditions help forecast the left tail of U.S. GDP, Plagborg-Møller et al. (2020) argue that no single financial predictor is persistently important for forecasting recessions. Their argument is based on the finding that the relationship between GDP and financial conditions is inherently unstable over time. This finding suggests that policy makers should not mechanically include financial conditions as predictors of tail events.

In order to understand the importance of global predictors in affecting country-level macroeconomic risks, we compare the QFAVAR with a restricted version that includes only the quantile factors and no global predictors $\boldsymbol{g}_t$. Consistent with the existing literature, we call this model the quantile dynamic factor model (QDFM). Fig. 3 plots the cumulative quantile score values for inflation and industrial production, computed using Eq. (13) in the previous subsection, for both the QFAVAR and QDFM models. Compared to average quantile scores, cumulative sums reflect the evolution of forecast performance of different models over the out-of-sample period. Given that the $QS_{t|t-h}(q)^M$ statistic is a loss function, the best model in each period is the one that maintains the lower cumulative values. The figure has four rows of nine panels, where the first (second) two rows correspond to the QS performance for the 10th and 90th percentile of inflation (industrial production) at horizon $h = 24$ and the columns correspond to the individual euro area countries.

Interesting patterns emerge from this graph. First, the QFAVAR has an overall lower quantile score for both tails than the QDFM in all panels. Second, in most panels, the largest reductions in loss compared to the QDFM occur in the middle of the sample. For the bottom tail of inflation the QFAVAR also gains compared to the QDFM early in the out-of-sample period and for industrial production at the beginning and towards the end of the sample for both tails. Accounting for the forecast horizon, these periods coincide with the euro area sovereign debt crisis, the Brexit referendum, and the Covid-19 pandemic; periods which were accompanied by strong comovements, heightened uncertainty, and global economic unrest. Overall, these results suggests that global variables help predict

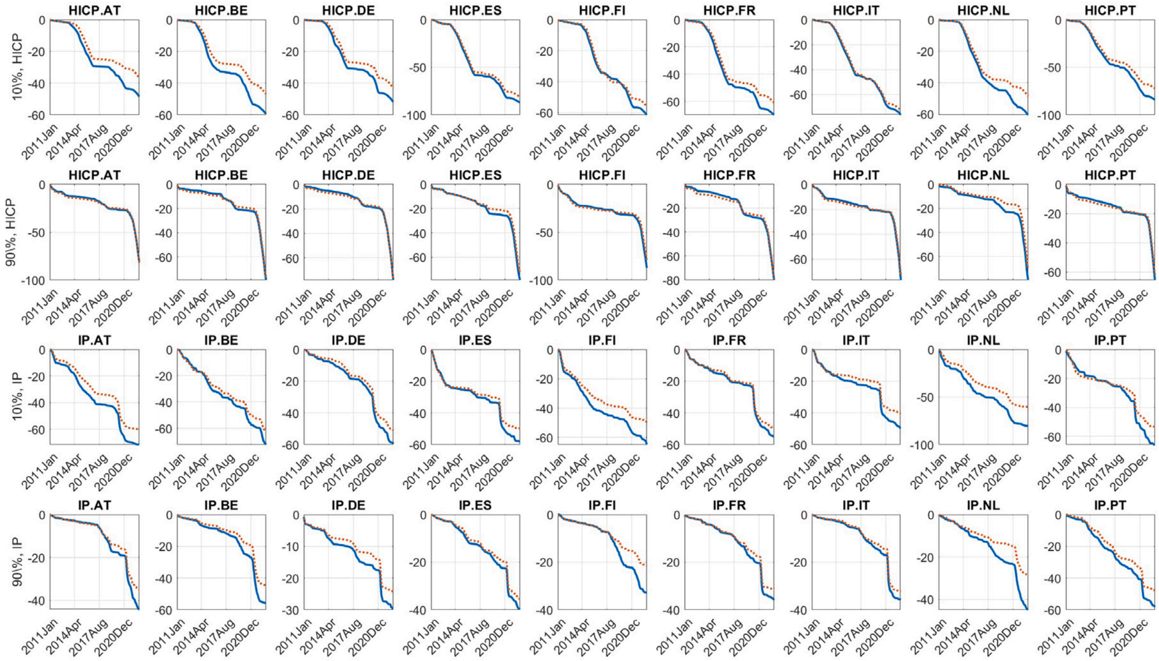

Fig. 3. Cumulative quantile score (QS) loss for forecast horizon $h = 24$. Lower values signify better performance. We compare two models: QFAVAR (blue solid line) and QDFM (orange dotted line). The out-of-sample evaluation period shown on the x-axis is 2011Jan to 2022Dec. The first two rows show quantile scores for the 10th and 90th percentiles of inflation, and the third and fourth rows show quantile scores for the 10th and 90th percentiles of industrial production.

tail risks in this multivariate multicountry setting, i.e. a setting with complex correlation structures. The conclusions of Plagborg-Møller et al. (2020) might hence only apply to univariate quantile models. For shorter forecast horizons the results are generally similar in that the QFAVAR gains relative to the QDFM in periods of heightened uncertainty. Additional supporting graphs for this exercise are provided in the Online Supplement.

4.3. Comparison with stochastic volatility models

Recent investigations by Brownlees and Souza (2021) and Carriero et al. (2023) indicate that VARs incorporating stochastic volatility (SV) and univariate Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models exhibit comparable performance to quantile regressions in forecasting macroeconomic tail risks. To examine whether this observation extends to our multivariate quantile regression model, we conduct a comparative analysis between the baseline QFAVAR and five benchmark SV regression models. The initial benchmark is a FAVAR with stochastic volatility in the state equation, akin to the approach by Mumtaz and Theodoridis (2017). Subsequently, we explore country-specific BVAR-SV models, following the methodology of Carriero et al. (2023). We also consider two crucial univariate benchmarks. The first is an autoregressive model of order two with stochastic volatility (AR(2)-SV), analogous to the GARCH model employed by Brownlees and Souza (2021). Finally, we examine the highly flexible Unobserved Components Stochastic Volatility (UCSV) model proposed by Stock and Watson (2007), acknowledged as one of the most effective reduced-form forecasting models in the literature.

It is worth noting that we can introduce stochastic volatility into the state equation of our QFAVAR model. A straightforward and parsimonious method is to allow the diagonal elements of the VAR covariance matrix, denoted as $\boldsymbol{\Omega}$, to vary over time. Details on modifying our algorithm to accommodate SV are available in Online Supplement Section S1.3, and we refer to this variant of our model as the QFAVAR-SV. Unlike SV in regression models, which captures time-varying uncertainty around the mean of a variable, the QFAVAR-SV models time-varying uncertainty at each quantile level. All models mentioned are estimated using MCMC algorithms, all of which are special cases of the algorithm for the QFAVAR. To ensure comparability, we employ automatic horseshoe priors and include two lags everywhere (except for the UCSV, where lag length selection is irrelevant). Additional details are available in the Online Supplement.

Analogous to the previous section, Fig. 4 plots the cumulative quantile score values for inflation and industrial production and forecast horizon $h = 24$. Overall, a few important features stand out. First, on average both, the QFAVAR and the QFAVAR-SV, dominate all benchmark models for the 10th percentile of inflation and industrial production. For the 90th percentile, the UC-SV is the best performing model, however, the QFAVAR and QFAVAR-SV outperform all other univariate and multivariate benchmarks. In the majority of panels, the largest relative gains emerge at the beginning, middle, and towards the end of the out-of-sample period. In line with the previous subsection, these dates coincide well with the euro area sovereign debt crisis, the Brexit referendum,

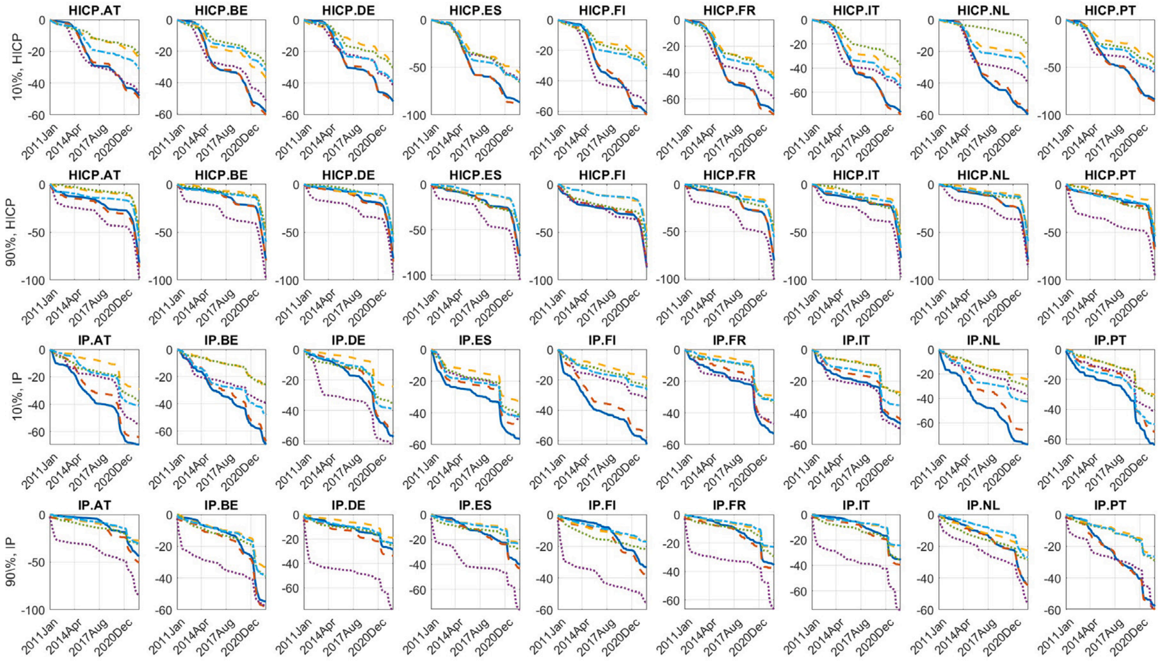

Fig. 4. Cumulative quantile score (QS) loss for forecast horizon $h = 24$. Lower values signify better performance. We compare six models: QFAVAR (blue solid line), QFAVAR-SV (orange dashed line), FAVAR-SV (yellow dashed line), UC-SV (purple dotted line), AR(2)-SV (green dotted line), and country-level VAR-SV (light blue dot-dashed line). The out-of-sample evaluation period shown on the x-axis is 2011Jan to 2022Dec. The first two rows show quantile scores for the 10th and 90th percentiles of inflation, and the third and fourth rows show quantile scores for the 10th and 90th percentiles of industrial production. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and the Covid-19 pandemic and suggest that the QFAVAR/QFAVAR-SV capture periods of strong comovements in uncertainty more accurately than competing models. Zooming in, in the majority of panels the QFAVAR and QFAVAR-SV perform very similarly, which is especially true for both tails of the inflation distribution. In addition, for the bottom tail of industrial production the QFAVAR clearly dominates the QFAVAR-SV throughout the sample. This suggests that allowing for stochastic volatility at best generates only marginal performance gains once quantile dynamics are accounted for. Comparing the individual benchmark models, no clear ranking emerges beyond the dominance of the UC-SV; however, in the majority of cases the country-level BVAR-SV provides more precise tail forecasts than the univariate benchmarks and the FAVAR-SV, which is estimated on the full set of data. Even though the structure of the FAVAR-SV is the same as the QFAVAR-SV, it has the worst quantile scores in the majority of cases.

Together, these findings provide some evidence that the joint quantile dynamics drive the performance gains of the QFAVAR/QFAVAR-SV for the most part. Specifically, the QFAVAR might be able to better capture the complex correlation structure of macroeconomic risk, especially in the bottom of the distribution. While this result is not necessarily at odds with Brownlees and Souza (2021) and Carriero et al. (2023), it suggest that allowing for quantile dynamics might lead to forecast performance gains compared to workhorse models such as BVAR-SVs, once a large-scale multivariate setting is considered. For shorter horizons, the QFAVAR and QFAVAR-SV on average outperform all benchmark models and predict tail risks more accurately. This result is especially clear for forecast horizons of one year or less. For $h = 1$, the FAVAR-SV and BVAR-SV emerge as the most competitive benchmark models for inflation and industrial production, respectively. However, the UC-SV remains dominant among the benchmark models for $h = 6, 12$. The full set of results is reported in the Online Supplement.

5. Global shocks and euro area macroeconomic risks

5.1. Illustrating heterogeneity

To get a first idea of which heterogeneities might emerge across the distribution of euro area data in practice, Fig. 5 presents generalized impulse response functions (GIRF) for the state equation, estimated with our QFAVAR. Even though GIRFs are not structural, they provide a great data-based tool to develop some guidance for designing our structural identification scheme in the next section. Each row of the figure depicts the GIRFs of the quantile factors to a change in one of the global variables. The last column presents the response of the global variable itself and the other columns contain the GIRFs for each block of variables, where the response of the 10th percentile factor is shown with dotted blue lines, the median with red solid lines, and that of the 90th percentile factor with yellow dashed lines. Because we extract the quantile factors per variable block, the factor level responses provide a measure of the response of the euro area aggregate.

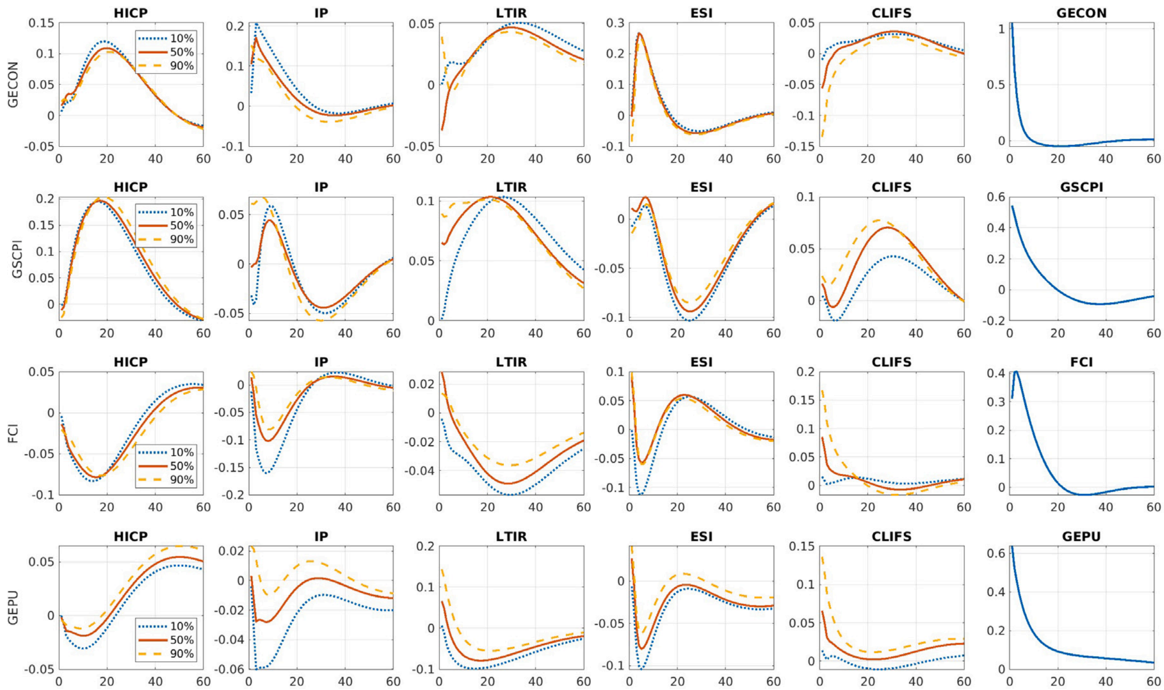

Fig. 5. Generalized impulse response functions from the QFAVAR. Each row represents shocks to one of the four global variables, while columns indicate the variable's response to each shock. The initial five columns present the posterior mean responses across the three quantile levels for the five quantile factors. The final column displays the posterior mean response for each corresponding global variable.

In general, the quantile IRF can be interpreted in two ways. On one hand, the individual responses at various quantile levels represent movements within different segments of the distribution. For instance, divergence of the responses of the 10th and 90th percentile factors reflects a widening of the distribution of the data and their responses to shocks. Additionally, a relatively stronger response of one factor compared to another can be indicative of skewness. As such, changes of the distribution imply changes to macroeconomic uncertainty and to the balance of upside and downside risks. Conversely, if all factor responses overlap, this implies a level shift in the distribution without changes to its shape or to uncertainty around the median response. Alternatively, the responses can also be interpreted through the lens of *scenario analysis*, i.e. as reflecting different states of the world. Depending on the economic environment, a policy maker might then pay particular attention to the IRF that represents the tail with the most adverse outcome as a benchmark case. This lends the IRFs an interpretation of "best" and "worst" case scenarios. Typically the left tail of output and both the left and right tails of inflation would be interpreted as worst-case scenarios for policy analysis.

At first glance, many different heterogeneities and responses emerge from Fig. 5 that can be broadly categorized into three groups. The first group contains e.g. the positive response of the ESI to an increase in the GECON or the positive response of HICP to an increase in the global supply chain pressure index, GSCPI. In both cases, the responses overlap for all quantile levels. Upside and downside risks around the median response hence remain balanced.

The second group contains impulse response functions that differ in magnitude but not in sign or shape across the different quantile levels. For example, the CLIFS quantile factors respond positively to an increase in financial conditions; however, the 90th percentile factor responds more strongly than the median factor which in turn responds more strongly than the 10th percentile factor. In this case, the 90th percentile factor provides a useful "worst-case" scenario, i.e. a situation where financial stress responds more than usual to a tightening of financial conditions as might be the case when financial markets are already in disarray. Conversely, the 10th percentile response could serve as a benchmark scenario when borrowing constraints and lending conditions are generally loose. Other examples include the negative (positive) response of the CLIFS factors to an increase in the GECON (GEPU).

The third group contains the most extreme forms of heterogeneities, i.e. quantile factor responses that differ strongly across quantile levels, in magnitude, shape, and sign. One example are the responses of the industrial production factors to changes in any of the global variables. An increase in the GECON for example coincides with positive on impact responses of all factors, but the 10th percentile factor responds by less than the median and 90th percentile, before the order reverses after about three to six months, when the 90th percentile responds the least and the 10th percentile factor the most. An increase in the GSCPI in turn leads to a strongly positive on impact response of the 90th percentile factor, a marginal response of the median factor, and a strongly negative on impact response of the 10th percentile factor. The 90th percentile response might hence serve as a benchmark when supply chain pressures are driven by strong demand rather than disruptions such as those observed during the pandemic. These might be better reflected by the response of the 10th percentile factor. For the responses to the FCI and the GEPU, strong heterogeneities emerge after about six months rather than on impact. In both cases the 10th percentile factor responds increasingly negatively compared

Table 3
Sign restrictions imposed in the QFAVAR for identifying global shocks.

	Global Shocks			
	Demand	Supply	Financial	Uncertainty
HICP factor (median)		+		0
IP factor (median)	+	0	−	−
LTIR factor (median)				
ESI factor (median)	+			−
CLIFS factor (median)			+	
10 and 90 percentile factors:	No sign or other restrictions			
GECON	1			
GSCPI		1		
FCI			1	
GEPU				1

to the other factors. On the one hand this suggests that an increase in economic policy uncertainty on average coincides with risks around industrial production tilting to the downside over the medium term. On the other hand, the more pronounced response of the 10th percentile might reflect economic environments where industrial production is particularly susceptible to policy uncertainty.

5.2. Sign restrictions in the QFAVAR

The results from the previous section have important implications for the design of structural identification schemes in the QFAVAR. Explicitly, because the QFAVAR contains different quantile levels of the same endogenous variable, restrictions across quantile levels for the same factors have to be imposed with care. Consider for example a QFAVAR in which zero restrictions are imposed across quantile levels for the same factor. In the previous section we observed that the on impact response of the quantile factors for the same variable are likely to be non-zero. If a tail factor is constrained not to respond on impact, this then induces spurious heterogeneity in the responses.

Similarly, one might consider applying the same sign restrictions commonly imposed to the mean/median, which are supported by economic theory, to the tail factors as well. This simplification would exclude cases for which the on impact responses of the quantile factors differ in sign. The previous section, however, suggests that such asymmetric behavior might be empirically relevant, and should not be excluded by theoretical restrictions. By imposing the same restrictions on both the median and the tails of a macroeconomic variable, important heterogeneities might be removed.

Incorporating these insights, sign, zero, and magnitude restrictions are imposed solely on the median factor, leaving the tail factors free of structural restrictions. This approach takes into account economic theory where guidance is available but does not constrain the rich dynamics that might be present in the tails. For instance, even with a zero restriction on the median, tail risks are explicitly allowed to vary on impact in response to the shock.

To illustrate these ideas, four global shocks are loosely identified using the QFAVAR: a global demand shock, a supply shock, a financial shock, and an uncertainty shock. A comprehensive overview of the restrictions is provided in Table 3.

The demand shock is identified by restricting the median IP factor and the median ESI factor to respond positively on impact. Similarly, for the supply shock, the median HICP factor is restricted to respond positively on impact, and the median factor of IP is restricted to zero, thereby potentially interpreting this shock as a global inflation shock. Downside risks to the industrial production outlook, such as a strongly negative response of the 10th percentile factor, are explicitly not ruled out as materializing in response to this shock.

For the financial shock, a negative sign is imposed on the response of the median industrial production factor, and a positive sign is imposed on the median CLIFS factor, resembling a shock to financial conditions or financial stress. As for the uncertainty shock, the on-impact response of the median HICP factor is restricted to zero, and negative on-impact responses are imposed on the median IP and ESI factor. The zero restriction does not, however, rule out on-impact responses of the tail factors of inflation, thereby allowing for changes in inflation risk.

Finally, the size of the response of GECON, GSCPI, FCI, and GEPU to the demand, supply, financial, and uncertainty shock, respectively, is constrained to unity. This is done for two reasons. First, it provides an easy way to standardize the scale of the shocks and impulse response functions. Second, identifying structural shocks in the global dimension is inherently challenging. This problem is exacerbated because all global variables in the QFAVAR are extracted from a set of economic variables spanning various categories and are, hence, prone to cross-contamination. Identification is therefore not sharp, and exclusion restrictions in the block of global variables are likely too restrictive

The impulse response functions under this identification scheme are given in Fig. 6, which is set up analogously to the figure in the previous section. In response to the demand shock, inflation, industrial production, consumer sentiment, and the interest rate increase, while financial stress declines. Despite the few restrictions, the footprint of this shock is hence indeed in line with a classical demand shock. However, the response of the individual quantile level factors is highly asymmetric, suggesting that the distribution of euro area financial stress skews to the right and that of all other variables to the left in response to the shock. Alternatively, the responses can be viewed through the lens of scenario analysis. If we believe domestic industrial production to increase little, but

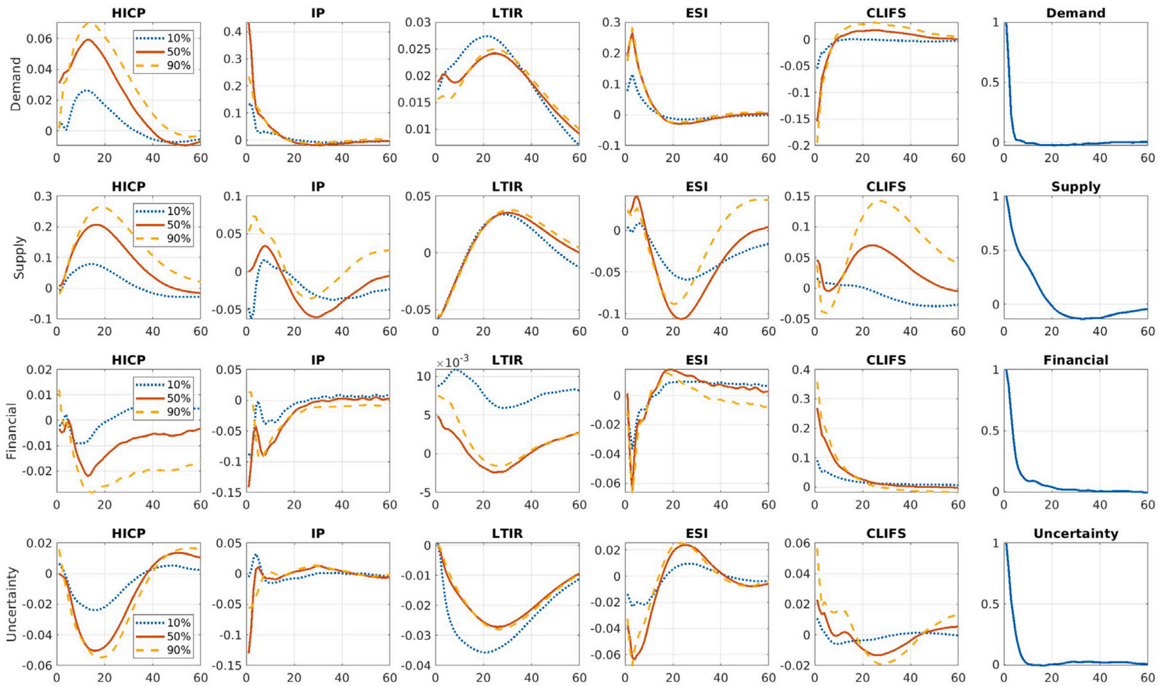

Fig. 6. Impulse response functions from the QFAVAR using the restrictions in Table 3. Each row represents shocks to one of the four global variables, while columns indicate the variable's response to each shock. The initial five columns present the posterior mean responses across the three quantile levels for the five factors. The final column displays the posterior mean response for each structural shock.

the pass through of global demand to inflation to be particularly strong in a given economic environment, we could exclusively consider the 90th percentile response of inflation and the 10th percentile response of IP as a benchmark.

For the supply shock, inflation responds little on impact. After about 6 months all factors respond positively and the distribution skews to the left. For industrial production, the median does not respond on impact due the zero restriction, but the 90th percentile responds positively and the 10th percentile negatively. Overall, uncertainty around IP is hence elevated in response to the shock. After about a year, the responses turn negative. This suggests possible scenarios in which inflation increases, but domestic industrial production either expands or contracts in response to the shock. Depending on the IRF considered, the responses might hence depend on the pass through of international price changes to the domestic economy.

In the event of a financial shock, for both, HICP and IP, the 90th percentile factor initially responds positively before all factors decline. The 90th percentile factor might hence reflect a scenario where the shock has a strong information component and financial conditions tighten in expectation of improving business cycle conditions and corresponding monetary policy. In both cases the 90th percentile and median decline by more than the 10th percentile, indicating risks that tilt to the downside over the medium term. The long term interest rate responds negatively in the median. However; the 10th percentile increases and the 90th percentile decreases in response to the shock. Uncertainty about future interest rates hence declines overall. The CLIFS react with the expected sign and financial stress generally increases. As before, the 90th percentile response of the CLIFS provides a benchmark when financial stress is believed to react particularly strongly to the shock, e.g. in a period of unrest in euro area financial markets.

Finally, an uncertainty shock is accompanied by a pronounced decline of IP on impact and at all percentile levels. HICP and the LTIR do not respond on impact, partially due to the zero restriction on the median of HICP, before responding strongly negatively in subsequent months. In most cases risks are increasingly on the downside. Both, the ESI and the CLIFS only respond little on impact, before assuming expected signs.

5.3. Forecast error variance decompositions

In this section, we investigate whether different structural shocks are particularly important for different parts of the data distribution and hence dominant drivers of tail risks for euro area aggregates. Fig. 7 presents forecast error variance decompositions for inflation and industrial production for the 10th percentile, the median, and the 90th percentile. The relative contributions of the demand, the supply, the financial, and the uncertainty shock are depicted in dark blue, light blue, green, and yellow, respectively.

For euro area inflation, the demand shock contributes the most and the policy uncertainty shock the least to the median on impact. For longer horizons, the supply shock becomes dominant, whereas all other shocks contribute roughly equal shares. This result differs markedly for the tails. In both tails, the uncertainty shock contributes about 40% on impact and hence the biggest share. The contribution of the demand and the financial shock are roughly equal. For longer horizons the supply shock contributes

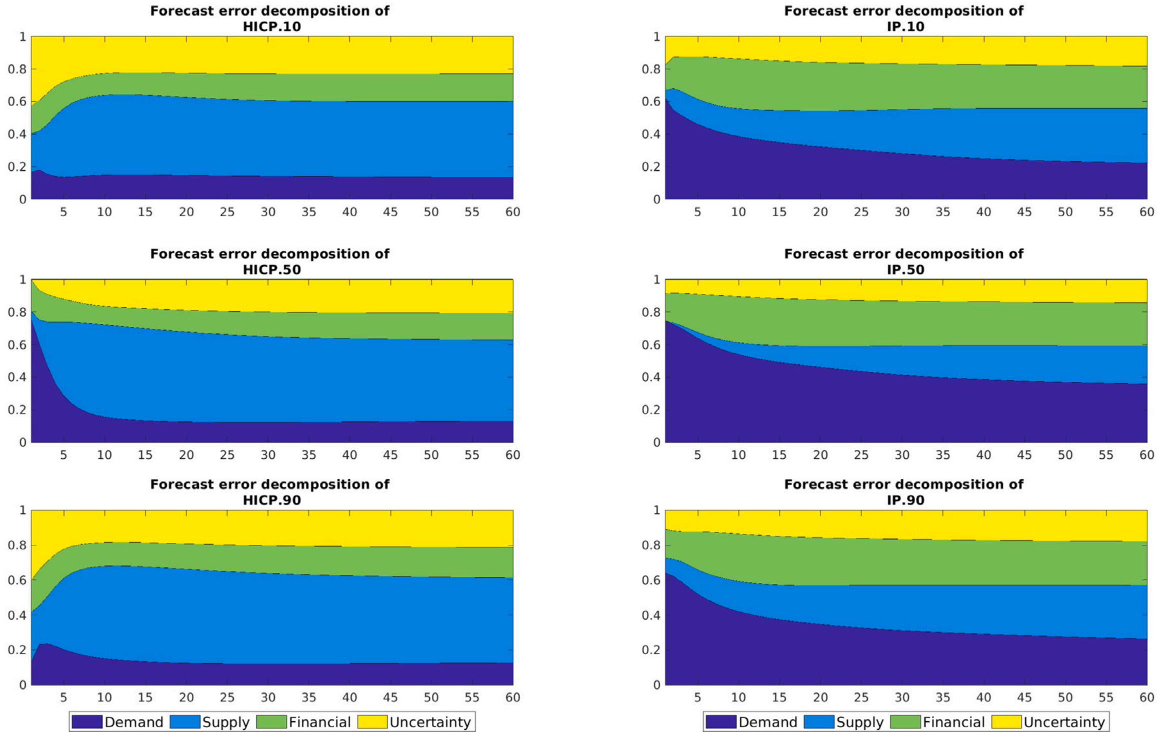

Fig. 7. Forecast error variance decomposition of the HICP and IP factors for the quantile levels $q = [0.1, 0.5, 0.9]$ and the 4 shocks considered in the structural exercise: Demand, Supply, Financial, and Uncertainty.

the most, the uncertainty shock the second most, and the demand shock is the smallest contributor. Overall, supply shocks and uncertainty shocks hence emerge as important drivers of tails risks for euro area inflation.

For industrial production, the results are more homogeneous across quantile levels. On impact the demand shock always contributes the most and the supply shock the least. At the median, the demand shock remains the dominant driver for longer horizons, followed by the financial shock and the supply shock. For both tails, the supply shock is the largest contributor and the uncertainty shock the smallest contributor for longer horizons. All other shocks contribute similar shares. Demand shocks hence emerge as the most important driver of short-run risks to industrial production, whereas supply shocks are particularly important for long-run tail risks.

5.4. Country-level responses

As explained in Section 2, the structure of the QFAVAR allows to project the impulse response functions for the quantile factors into the space of the individual country level variables via the measurement equation. This enables the study of heterogeneities at the country level and allows to uncover further cross-sectional heterogeneity. A key ingredient is the matrix $\boldsymbol{\Gamma}$ that contains the parameters for the global variables in the measurement equation. If $\boldsymbol{\Gamma} = \mathbf{0}$, then the global variables do not enter the measurement equation directly. In this case, the country level IRF will be mirror-images of the factor level IRF up to changes in scale. If $\boldsymbol{\Gamma} \neq \mathbf{0}$, then the global variables impact the response of the country level data directly. Depending on the exposure of to the global variables, the country level responses then differ beyond changes in scale. A similar logic applies to any other predictor that one might want to include in the measurement equation.

As an example of strong cross sectional heterogeneities, Fig. 8 illustrates the country level IRF for Germany, France, Spain, and Italy to an uncertainty shock. The full set of country level responses to all shocks is provided in the Online Supplement. On impact of the shock, the quantiles for inflation in Germany and France increase, before contracting and reaching their trough after about 12 months. In both countries, this contraction is more pronounced for the 90th percentile than the median and the 10th percentile, respectively. These results differ strongly from the responses in Spain and Italy. In both countries, the 10th percentile and the median contract on impact, where the contraction is more pronounced for the latter. In general, the scale of this contraction is also larger than in France and Germany.

The results for industrial production are more homogeneous. In all four countries, most quantile levels contract in response to the shock. Differences emerge for the scale of the response of IP which is more marked for the median in Germany, and to a lesser extent in France, compared to other countries.

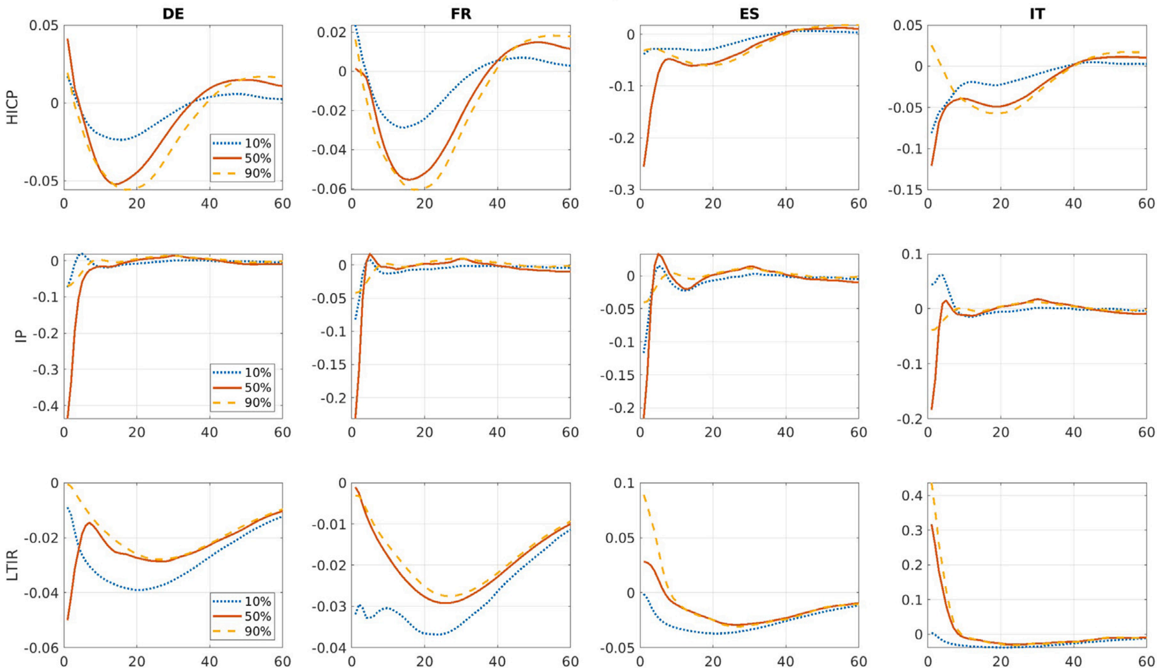

Fig. 8. Impulse response functions from the QFAVAR using the restrictions in Table 3. Each row represents the response of HICP, IP, and LTIR to an uncertainty shock. The columns indicate a selected country.

Finally, for the long-term interest rate, two main differences stand out. First, the quantiles for Germany and France decline in response to the shock, reaching their lowest value after about 24 months. In Italy and Spain, the quantiles increase on impact. For Italy the responses remain positive, whereas they become slightly negative for Spain after 12 months.

Overall, these results might point towards heterogeneity in response to an uncertainty shock across euro area countries or groups of countries such as the core countries Germany and France and periphery countries Italy and Spain. Because divergent quantile responses across countries and hence in the build-up of economic uncertainty might lead to fragmentation across the monetary union, these differences are particularly important from a euro area monetary policy perspective.

6. Conclusions

We develop a new quantile factor-augmented vector autoregressive (QFAVAR) model that is a natural extension of the popular FAVAR for targeting specific quantiles of the distribution of macroeconomic data. The advantage of the factor-based approach over quantile VAR modeling is that it is both flexible and parsimonious. The factors not only summarize cross-sectional correlations across different countries, but also across different quantiles of the data distribution. Using a Bayesian perspective, estimation of the QFAVAR adds only a minor level of complexity relative to a Markov chain Monte Carlo algorithm for the classical FAVAR model (Bernanke et al., 2005).

The proposed QFAVAR is fully parametric (likelihood-based), meaning that our proposed specification can easily extend to incorporate other formulations without inducing huge estimation and setup costs. First, we can obtain quantile dynamic factor models and univariate quantile autoregressions as special cases of the QFAVAR, simply by restricting certain parameters of the state–space form of the model. Second, we can incorporate interesting features such as time-varying parameters and stochastic volatility in the measurement and/or state equations of the model in order to allow for more flexible inference. Finally, we show how structural analysis can be conducted with the QFAVAR. We demonstrate this in an empirical exercise in which we identify global demand, supply, financial, and uncertainty shocks using sign, zero, and magnitude restrictions. The corresponding impulse response functions reveal strong asymmetries in the response of macroeconomic risk around euro area aggregates and country level data in response to these shocks.

Appendix A. Technical appendix: Derivation of the linear state–space form

Throughout the following, lower case letters indicate scalars, bold lower case letters indicate vectors, and bold upper case letters denote matrices. Let $y_{ij,t}$ denote a macroeconomic or financial variable $i = 1, \ldots, m$ for country $j = 1, \ldots, n$ that is observed for time

$t = 1, \dots, T$. Additionally, let $q \in (0,1)$ denote a given quantile level. The quantile factor regression for variable $y_{ij,t}$ for quantile level q is of the form

$$y_{ijt} = c_{ij(q)} + \lambda_{ij(q)} f^{i}_{t(q)} + \boldsymbol{\gamma}_{ij(q)} \boldsymbol{g}_t + u_{ij,t(q)}, \quad u_{ij,t(q)} \sim AL(0, \sigma_{ij(q)}, q) \tag{A.1}$$

Stacking all variables over i, j, $i = 1, \dots, m$, $j = 1, \dots, n$, into the vector $\boldsymbol{y}_t = \left[y_{11t}, \dots, y_{1nt}, \dots, y_{m1t}, \dots, y_{mnt} \right]'$ we obtain the model

$$\boldsymbol{y}_t = \boldsymbol{c}_q + \boldsymbol{\lambda}_{(q)} \boldsymbol{f}_{t(q)} + \boldsymbol{\gamma}_{(q)} \boldsymbol{g}_t + \boldsymbol{u}_{t(q)}, \tag{A.2}$$

where $\boldsymbol{c}_{(q)} = \left[c_{11(q)}, \dots, c_{1n(q)}, \dots, c_{m1(q)}, \dots, c_{mn(q)} \right]'$ is an nm×1 vector, $\boldsymbol{\gamma}_{(q)} = \left[\boldsymbol{\gamma}'_{11(q)}, \dots, \boldsymbol{\gamma}'_{1n(q)}, \dots, \boldsymbol{\gamma}'_{m1(q)}, \dots, \boldsymbol{\gamma}'_{mn(q)} \right]'$ is an nm×k matrix, $\boldsymbol{u}_{t(q)} = \left[u_{11t(q)}, \dots, u_{1nt(q)}, \dots, u_{m1t(q)}, \dots, u_{mnt(q)} \right]'$ is an nm×1 vector,

$$\boldsymbol{\Lambda}_{(q)} = blkdiag\left(\boldsymbol{\lambda}_{1(q)}, \boldsymbol{\lambda}_{2(q)}, \dots, \boldsymbol{\lambda}_{m(q)} \right) = \begin{bmatrix} \begin{bmatrix} \lambda_{11(q)} \\ \vdots \\ \lambda_{1n(q)} \end{bmatrix} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \begin{bmatrix} \lambda_{21(q)} \\ \vdots \\ \lambda_{2n(q)} \end{bmatrix} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \mathbf{0} \\ \mathbf{0} & \dots & \mathbf{0} & \begin{bmatrix} \lambda_{m1(q)} \\ \vdots \\ \lambda_{mn(q)} \end{bmatrix} \end{bmatrix}$$

and $\boldsymbol{f}_{t(q)} = \begin{bmatrix} f^1_{t(q)} \\ f^2_{t(q)} \\ \vdots \\ f^m_{t(q)} \end{bmatrix}$. Eq. (A.2) shows the FAVAR measurement equation for each quantile level q. Stacking across r quantiles $q = q_1, \dots, q_r$ we obtain

$$\boldsymbol{Y}_t = \boldsymbol{c} + \boldsymbol{\Lambda} \boldsymbol{F}_t + \boldsymbol{\Gamma} \boldsymbol{g}_t + \boldsymbol{u}_t, \tag{A.3}$$

where $\boldsymbol{c} = \left[\boldsymbol{c}'_{(q_1)}, \dots, \boldsymbol{c}'_{(q_r)} \right]'$ is an $(nmr \times 1)$ vector, $\boldsymbol{\Gamma} = \left[\boldsymbol{\gamma}'_{(q_1)}, \dots, \boldsymbol{\gamma}'_{(q_r)} \right]'$ is an $(nmr \times k)$ matrix, $\boldsymbol{F}_t = \left[\boldsymbol{f}'_{t(q_1)}, \dots, \boldsymbol{f}'_{t(q_r)} \right]'$ is an $(mr \times 1)$ vector and $\boldsymbol{\Lambda} = blkdiag\left(\boldsymbol{\Lambda}_{(q_1)}, \dots, \boldsymbol{\Lambda}_{(q_r)} \right)$ is an $(nmr \times mr)$ matrix.

Augmenting the measurement Eq. (A.3) with an identity for $\boldsymbol{g}_t$ and combining it with the state Eq. (4) we obtain the state–space form of the QFAVAR

$$\begin{bmatrix} \boldsymbol{Y}_t \\ \boldsymbol{g}_t \end{bmatrix} = \boldsymbol{c} + \begin{bmatrix} \boldsymbol{\Lambda} & \boldsymbol{\Gamma} \\ \mathbf{0} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{F}_t \\ \boldsymbol{g}_t \end{bmatrix} + \begin{bmatrix} \boldsymbol{u}_t \\ \mathbf{0} \end{bmatrix}, \tag{A.4}$$

$$\begin{bmatrix} \boldsymbol{F}_t \\ \boldsymbol{g}_t \end{bmatrix} = \boldsymbol{v} + \boldsymbol{\Phi}_1 \begin{bmatrix} \boldsymbol{F}_{t-1} \\ \boldsymbol{g}_{t-1} \end{bmatrix} + \dots + \boldsymbol{\Phi}_p \begin{bmatrix} \boldsymbol{F}_{t-p} \\ \boldsymbol{g}_{t-p} \end{bmatrix} + \boldsymbol{\varepsilon}_t. \tag{A.5}$$

Each element of the vector $\boldsymbol{u}_t$ is distributed as independent univariate asymmetric Laplace. As explained in the main text we can write the asymmetric Laplace as a Gaussian-Exponential location-scale mixture, in which case we the state–space model above is in conditionally normal form and sampling of the state-vector $\left[\boldsymbol{F}'_t, \boldsymbol{g}'_t \right]'$ is feasible using the simulation smoother of Carter and Kohn (1994). Finally, note that sampling of the state-form requires first-order Markov dependence of the state variable. In Eq. (A.5) above the state vector follows a VAR(p) but we can use standard tools for writing it in VAR(1) companion form, (see Lütkepohl, 2005, Chapter 2). Detailed derivations of MCMC algorithms for inference, are provided in the Online Supplement.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jeconom.2024.105730.

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